Multiple Choice:

1. In an n-p-n transistor, the majority carriers in the base are

   A. electrons
   B. holes
   C. both holes and electrons
   D. either holes or electrons

2. For a transistor to function as an amplifier,

   A. both the EB and CB junctions must be forward-biased.
   B. both the EB and CB junctions must be reverse-biased.
   C. the EB junction must be forward-biased and the CB junction must be-biased.
   D. the CB junction must be forward-biased and the EB junction must be reverse-biased.

3. Voltage-divider bias provides:

   A. an unstable Q point
   B. a stable Q point
   C. a Q point that easily varies with changes in the transistor's current gain
   D. a Q point that is stable and easily varies with changes in the transistor's current gain

4. A bipolar junction transistor operates with a collector current of 1.2 A and a base current of 50 mA. What will the value of emitter current be?

   A. 1.25 A
   B. 0.05 A
   C. 1.2 A
   D. 1.15 A

5. On the schematic symbol of a pnp transistor,

   A. the arrow points out on the emitter lead.
   B. the arrow points out on the collector lead.
   C. the arrow points in on the base lead.
   D. the arrow points in on the emitter lead.

6. In a transistor, which current is the largest?

   A. I_C.
   B. I_E.
   C. I_B.
   D. I_D.

7. Determine R_B and R_C for a fixed-bias configuration if V_{CC} = 12 V, β= 80, and I_{CQ} = 2.5 mA with V_{CEQ} =6 V. Use standard values.

   A. 361.6 kΩ ; 2.4 kΩ
   B. 361.6 kΩ ; 4.2 kΩ
   C. 316.6 kΩ ; 2.4 kΩ
   D. 316.6 kΩ ; 4.2 kΩ

8. To operate properly, a transistor's base-emitter junction must be forward biased with reverse bias applied to which junction?

   A. collector-emitter
   B. base-collector
   C. base-emitter
   D. collector-base

9. For a certain BJT, the following values are given, β = 60, I_{ceo}=1μA and I_c =1.5 mA. Compute for the value of I_E.

   A. 1.67 mA
   B. 1.23 mA
   C. 1.89 mA
   D. 1.52 mA

10. The primary function of the bias circuit is to

    A. hold the circuit stable at V_{CC}
    B. hold the circuit stable at V_{in}
    C. ensure proper gain is achieved
    D. hold the circuit stable at the designed Q-point

11. A class C amplifier is driven by a 200 kHz signal. The transistor is on for 1 µs, and the amplifier is operating over 100 %of its load. If $I_{C(sat)}$ = 100 mA and $V_{CE(sat)}$ = 0.2 V. what is the average power dissipation?

    A. 1 mW
    B. 4 mW
    C. 3 mW
    D. 5 mW

12. For a typical transistor, which two currents are nearly the same?

    A. I_B and I_E.
    B. I_B and I_C.
    C. I_C and I_E.
    D. None of the above.

13. The ends of a load line drawn on a family of curves determine:

    A. saturation and cutoff
    B. the operating point
    C. the power curve
    D. the amplification factor

14. Which of the following transistor amplifier configurations provides a 180° phase shift between the ac input and output voltages?

    A. the common-emitter amplifier.
    B. the emitter follower.
    C. the common-base amplifier.
    D. the common-collector amplifier.

15. Which region in a transistor is the most heavily region?

    A. the emitter region.
    B. the collector region.
    C. the gate region.
    D. the base region.

16. Determine I_c if I_e = 2.8 mA and I_B = 20 uA.

    A. 2.78 mA
    B. 2.87 mA
    C. 2.14 mA
    D. 2.41 mA

17. In a bipolar transistor the barrier potential

    A. 0
    B. a total of 0.7 V
    C. 0.7 V across each depletion layer
    D. 0.35 V

18. Which transistor region is very thin and lightly doped?

    A. the emitter region.
    B. the collector region.
    C. the anode region.
    D. the base region.

19. A swamping resistor in a common-emitter amplifier

    A. stabilizes the voltages gain.

B. increases the voltage gain.
C. reduces distortion.
D. Both A and C.

20. Which of the following biasing techniques produces the most unstable Q point?

A. voltage divider bias.
B. emitter bias.
C. collector bias.
D. base bias.

21. When a transistor is in saturation.

A. $V_{CE} = V_{CC}$.
B. $I_C = 0$ A.
C. $V_{CE} = 0$ V.
D. $V_{CE} = ½ V_{CC}$.

22. With voltage divider bias, how much is the collector-emitter voltage, V_{CE}, when the transistor is cut off?

A. $V_{CE} = 1.2 V_{CC}$.
B. $V_{CE} = V_{CC}$.
C. $V_{CE} = 0$ V.
D. None of the above.

23. Given an α_{dc} of 0.998, determine I_c if I_e = 4 mA.

A. 2.992 mA
B. 3.992 mA
C. 4.992 mA
D. 5.992 mA

24. In a class B push-pull amplifier, the transistors are biased slightly above cutoff to avoid

A. crossover distortion
B. unusually high efficiency
C. negative feedback
D. a low input impedance

25. Which of the following transistor amplifier configurations is often used in impedance-matching applications?

A. The common-base amplifier
B. The emitter follower
C. The common-emitter amplifier
D. The common-source amplifier

26. In a C-E configuration, an emitter resistor is used for:

A. stabilization
B. ac signal bypass
C. collector bias
D. higher gain

27. A transistor is biased so that I_{BQ} = 120 µA, the following values are given, I_{CBQ} = 10 µA and α = 0.96. Find the value of I_{EQ}.

A. 3.25 mA C. 3.15 mA
B. 3.20 mA D. 3.10 mA

28. What is the biggest disadvantage of the common-base amplifier?

A. its high input impedance.
B. its low voltage gain.
C. its high output impedance.
D. its low input impedance.

29. In an n-p-n transistor biased for operation in forward active region

A. emitter is positive with respect to base
B. collector is positive with respect to base
C. base is positive with respect to emitter and collector is positive with respect to base
D. none of the above

30. In what operating region does the collector of a transistor act like a current source?

A. the active region.
B. the saturation region.
C. the cutoff region.
D. the breakdown region.

31. In a transistor, which is the largest of all the doped regions?

A. the emitter region.
B. the collector region.
C. the gate region.
D. the base region.

32. What is the only amplifier configuration that provides both voltage and current gains?

A. The common-base amplifier
B. The emitter follower
C. The common-emitter amplifier
D. The common-collector amplifier

33. In a common-emitter amplifier, what happens to the voltage gain, Av, when a load resistor, R_L, is connected to the output?

A. A_v decreases.
B. A_v increases.
C. A_v doesn't change.
D. A_v doubles

34. In which type of amplifier is the input applied to the emitter and the output taken from the collector?

A. The common-base amplifier
B. The common-emitter amplifier
C. The common-collector amplifier
D. The emitter follower

35. Which type of transistor amplifier is also known as the emitter follower?

A. the common-base amplifier.
B. the common-collector amplifier.
C. the common-emitter amplifier.
D. None of the above.

36. In which region of a CE bipolar transistor is collector current almost constant?

A. Saturation region
B. Active region
C. Breakdown region
D. Both saturation and active region

37. The endpoints of a dc load line are labeled

A. I_{CQ} and V_{CEQ}.
B. $I_{C(Sat)}$ and $V_{CE(off)}$.
C. $I_{C(off)}$ and $V_{CE(Sat)}$.
D. None of the above.

38. Which of the following conditions are needed to properly bias an npn transistor amplifier?

A. Forward bias the base/emitter junction and reverse bias the base/collector junction.
B. Forward bias the collector/base junction and reverse bias the emitter/base junction.

C. Apply a positive voltage on the n-type material and a negative voltage on the p-type material.
D. Apply a large voltage on the base.

39. The α_{dc} of a transistor equals

A. I_C/L_E.
B. I_E/L_C.
C. I_B/L_C.
D. I_C/I_B.

40. In which type of amplifier is the input applied to the emitter and the output taken from the collector?

A. The common-base amplifier
B. The common-emitter amplifier
C. The common-collector amplifier
D. The emitter follower

41. In a transistor amplifier, what happens to the collector voltage, V_c, when the collector current, I_c increases?

A. V_c increase.
B. V_c stays the same.
C. V_c decreases.
D. It cannot be determined.

42. When the collector current in a transistor is zero, the transistor is

A. cut off.
B. saturated.
C. operating in the breakdown region.
D. Either B or C.

43. Removing the emitter bypass capacitor in a common-emitter amplifier will

A. decrease the voltage gain, A_v.
B. increase the input impedance, Z_{in}.
C. increase the voltage gain, A_v.
D. Both A and B.

44. With the positive probe on an NPN base, an ohmmeter reading between the other transistor terminals should be:

A. open
B. infinite
C. low resistance
D. high resistance

45. A bipolar junction transistor has

A. only one p-n junction.
B. three p-n junctions.
C. no p-n junctions.
D. two p-n junctions.

46. Emitter bias with two power supplies provides a

A. very unstable Q point.
B. very stable Q point.
C. large base voltage.
D. None of the above.

47. For a transistor operating in the active region,

A. $I_C = \beta_{dc} X I_B$.
B. V_{cc} has little or no effect on the value of I_C.
C. I_C is controlled solely by V_{CC}.
D. Both A and B.

48. In a bipolar transistor which current is largest

A. collector current
B. base current
C. emitter current
D. base current or emitter current

49. Which of the following transistor amplifiers has the lowest output impedance?

A. the common-collector amplifier.
B. the common-base amplifier.
C. the common-emitter amplifier.
D. None of the above.

50. The power dissipation in a transistor is the product of

A. emitter current and emitter to base voltage
B. emitter current and collector to emitter voltage
C. collector current and collector to emitter voltage
D. none of the above

END